

Exercise 6



Amirreza "Farnam" Taheri

Student ID: 4021300198003

Macroeconomics I

Instructor: Prof. Hosein Joshaghani

May 1, 2024

1.

(Household's Problem)

The representative household maximizes its lifetime utility function subject to its budget constraint:

$$\max U = \mathbb{E}_t \left[\sum_{i=0}^{\infty} \beta^{t+i} \left(\frac{C_{t+i}^{1-\sigma}}{1-\sigma} - \frac{N_{t+i}^{1+\eta}}{1+\eta} \right) \right] \quad (1)$$

$$\text{s.t.} \quad C_t + i_t = W_t N_t + R_t K_t \quad (2)$$

Substituting the law of motion for capital, $K_{t+1} = (1 - \delta)K_t + i_t$, into the budget constraint:

$$C_t + K_{t+1} = W_t N_t + (1 + R_t - \delta)K_t \quad (3)$$

The Lagrangian for firm problem is:

$$\mathcal{L} = E_t \sum_{i=0}^{\infty} \beta^{t+i} \left[\left(\frac{C_{t+i}^{1-\sigma}}{1-\sigma} - \frac{N_{t+i}^{1+\eta}}{1+\eta} \right) - \mu_{t+i} (C_t + K_{t+1} - W_t N_t - (1 + R_t - \delta)K_t) \right] \quad (4)$$

The FOCs are:

$$\frac{\partial \mathcal{L}}{\partial C_{t+i}} = C_{t+i}^{-\sigma} - \mu_{t+i} = 0 \quad (5)$$

$$\frac{\partial \mathcal{L}}{\partial N_{t+i}} = N_{t+i}^{\eta} - \mu_{t+i} = 0 \quad (6)$$

$$\frac{\partial \mathcal{L}}{\partial K_{t+i}} = \beta \mathbb{E} \mu_{t+i+1} (1 + R_t - \delta) - \mu_{t+i} = 0 \quad (7)$$

The intertemporal optimality conditions are: Substituting equation (7) into the equation (9), we get:

$$\beta \mathbb{E} C_{t+i+1}^{-\sigma} (1 + R_t - \delta) = C_{t+i}^{-\sigma} \quad (8)$$

Similarly from equation (8) and (9), we get:

$$\beta \mathbb{E} N_{t+i+1}^{\eta} (1 + R_t - \delta) = N_{t+i}^{\eta} \quad (9)$$

The intratemporal optimality condition in the equilibrium:

$$\frac{N_{t+i}^{\eta}}{C_{t+i}^{-\sigma}} = W_{t+i} \quad (10)$$

(Firm's Problem)

To show that firms won't make a profit in this economy, we assume perfect competition in the factor markets, which implies that firms take prices as given and thus price equals marginal cost. In equilibrium, firms make zero economic profit.

The firm's technology is described by the production function:

$$Y_t = z_t K_t^\alpha N_t^{1-\alpha} \quad (1)$$

The technology follows an *AR1* process:

$$\log(z_{t+1}) = \rho \log(z_t) + \epsilon_t, \quad \epsilon_t \sim IID \quad (2)$$

Firm wants to maximize its profit:

$$\max \quad \Pi_t = z_t K_t^\alpha N_t^{1-\alpha} - W_t N_t - R_t K_t \quad (3)$$

The FOCs are:

$$\frac{\partial \Pi}{\partial N_t} = (1 - \alpha) z_t K_t^\alpha N_t^{-\alpha} - W_t = 0 \quad (4)$$

$$\frac{\partial \Pi}{\partial K_t} = \alpha z_t K_t^{\alpha-1} N_t^{1-\alpha} - R_t = 0 \quad (5)$$

Plugging the optimal values into the Firm's profit we will see that firm's profit is equal to zero.

(The Steady State)

The steady states equations are as:

$$K_{t+1} = (1 - \delta)K_t + i_t = (1 - \delta)K_t + z_t K_t^\alpha N_t^{1-\alpha} - C_t \quad (1)$$

$$\beta \mathbb{E} C_{t+1}^{-\sigma} (1 + R_t - \delta) = C_t^{-\sigma} \quad (2)$$

$$\frac{N_t^\eta}{C_t^{-\sigma}} = W_t = (1 - \alpha) z_t K_t^\alpha N_t^{-\alpha} \quad (3)$$

In the Steady State, all variables are constant over time, and the economy is in a state of long-run equilibrium. The Steady State equations describe the relationships between the variables in this equilibrium state. In the steady state we have no noise, assuming $z^* = 1$, we have:

$$K^* = (1 - \delta)K^* + i^* = (1 - \delta)K^* + K^{*\alpha} N^{*1-\alpha} - C^* \quad (4)$$

$$\beta \mathbb{E} C^{*-\sigma} (1 + R_t - \delta) = C^{*-\sigma} \quad (5)$$

$$\frac{N^{*\eta}}{C^{*-\sigma}} = W^* = (1 - \alpha) K^{*\alpha} N^{*-\alpha} \quad (6)$$

2. Intuition

For the steady states equations, the intuitions are as:

Capital Accumulation:

The steady state equation for capital accumulation states that the change in capital is equal to investment minus depreciation.

Consumption:

The steady state equation for consumption states that consumption is equal to output minus investment.

Labor:

The steady state equation for labor states that labor is equal to the marginal product of labor.

Note that equation of the previous sections are already explained in the respective section.